

STUDENT PERFORMANCE TRACKER USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

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ABSTRACT

The Student Performance Tracker is a Machine Learning project designed to analyze and predict student academic performance. The system uses factors such as study hours, attendance percentage, and assignment completion to estimate student marks. A Linear Regression algorithm is employed to build the predictive model. The project also provides performance evaluation metrics, graphical visualizations, and student performance categorization, helping educators monitor and improve academic outcomes.

INTRODUCTION

Educational institutions continuously seek effective methods to evaluate and improve student performance. Traditional evaluation techniques often rely solely on examination results, which may not provide a complete understanding of student progress.

This project utilizes Machine Learning techniques to analyze various academic factors and predict student performance. By using historical data and predictive analytics, educators can identify strengths and weaknesses in student learning behavior and take appropriate actions.

OBJECTIVES

- To track and analyze student academic performance.
 - To predict student marks using Machine Learning techniques.
 - To evaluate the impact of study hours, attendance, and assignments on academic performance.
 - To generate visual insights through data analysis.
 - To assist educators in identifying students requiring academic support.
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TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 500 MB free space

Software Requirements

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas Library
- NumPy Library
- Matplotlib Library
- Scikit-learn Library

DATASET DESCRIPTION

The dataset contains academic information of students with the following attributes:

Attribute	Description
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Student_ID	Unique identification number
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Study_Hours	Number of hours spent studying
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Attendance	Attendance percentage
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Assignments	Number of assignments completed
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Marks	Student examination marks
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The dataset consists of 20 student records used for training and testing the model.

METHODOLOGY

Step 1: Data Collection

A student academic dataset was created containing study hours, attendance, assignments, and marks.

Step 2: Data Preprocessing

The dataset was organized using Pandas DataFrames and checked for consistency.

Step 3: Feature Selection

The following features were selected:

- Study Hours
- Attendance
- Assignments

Target Variable:

- Marks

Step 4: Data Splitting

The dataset was divided into training and testing datasets using the train-test split technique.

Step 5: Model Training

A Linear Regression model was trained using the selected features.

Step 6: Model Evaluation

The model performance was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Step 7: Prediction

The model predicts marks for a new student based on academic factors.

Step 8: Visualization

Various charts were generated to analyze:

- Study Hours vs Marks
 - Attendance vs Marks
 - Student Marks Analysis
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IMPLEMENTATION

The project was implemented using Python programming language. The dataset was loaded into a Pandas DataFrame and processed for machine learning analysis. A Linear Regression model was trained on the dataset and used to predict student marks.

The project also calculates evaluation metrics and displays graphical visualizations to better understand academic trends and performance patterns.

RESULTS

The system successfully:

- Predicted student marks accurately.
- Evaluated model performance using standard metrics.
- Categorized student performance levels.
- Generated visual insights into academic performance.
- Identified top-performing students.

Performance Categories:

- Excellent
 - Very Good
 - Good
 - Average
 - Needs Improvement
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ADVANTAGES

- Easy monitoring of student performance.
 - Accurate prediction of academic results.
 - Supports data-driven educational decisions.
 - Provides graphical analysis of student progress.
 - Helps identify students needing improvement.
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APPLICATIONS

- Schools
- Colleges
- Universities

- Online Learning Platforms
 - Academic Performance Monitoring Systems
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FUTURE ENHANCEMENTS

- Integration with real-world student databases.
 - Development of a web-based dashboard.
 - Real-time performance tracking.
 - Advanced machine learning algorithms.
 - Automated report generation and notifications.
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CONCLUSION

The Student Performance Tracker successfully demonstrates the application of Machine Learning in the education sector. By analyzing factors such as study hours, attendance, and assignments, the system can accurately predict student marks and performance levels. The project provides valuable insights that can assist educators in improving student outcomes and academic planning.

REFERENCES

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 2. Pandas Documentation
 3. NumPy Documentation
 4. Matplotlib Documentation
 5. Scikit-learn Documentation
 6. Machine Learning Research Papers
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